Week 2
Lecture 3
The Process of Interaction Design

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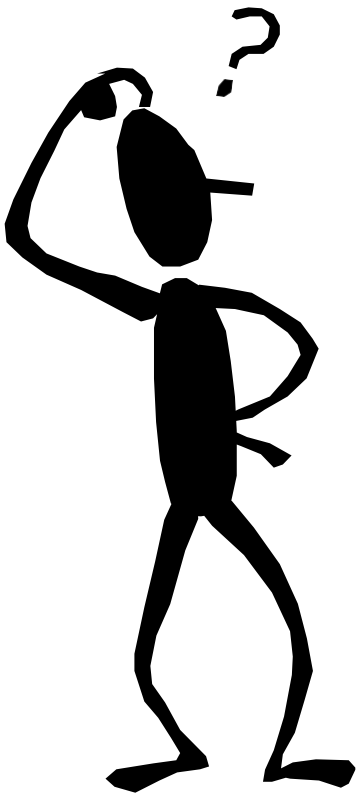
Overview

- What is involved in Interaction Design?
  - Understanding the problem space
  - Importance of involving users
  - Degrees of user involvement
  - What is a user-centered approach?
  - Four basic activities of interaction design



What is involved in interaction design?

It is a process:

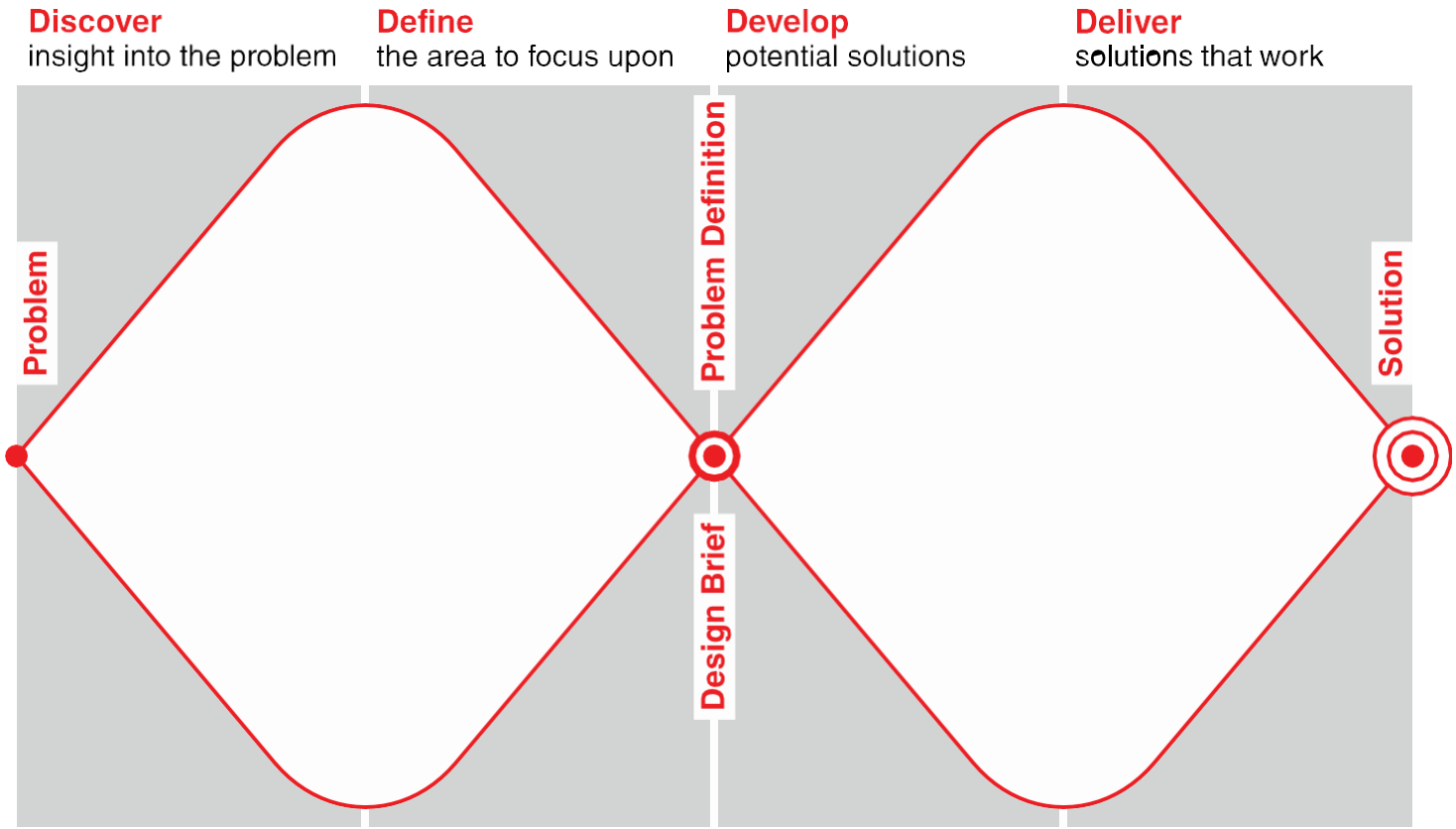
- Focused on **discovering requirements**, **designing** to fulfil requirements, **producing prototypes** and **evaluating** them
- Focused on users and their goals
- Involves trade-offs to balance conflicting requirements
- Generating alternatives and choosing between them is key

What is involved in interaction design?

Different approaches:

- User-centered design,
- Activity-centered design,
- Systems design.

The Double Diamond of Design



https://www.uxpin.com/studio/blog/double-diamond-design-process/

Understanding the problem space

Explore

- What is the current user experience?
- Why is a change needed?
- How will this change improve the situation?

Importance of involving users

Expectation management

- Realistic expectations
- No surprises, no disappointments
- Timely training
- Communication, but no hype

Ownership

- Make the users active stakeholders
- More likely to forgive or accept problems
- Can make a big difference in acceptance and success of product

Degrees of user involvement

- Member of the design team
  - Fulltime: continuous input, but lose touch with other users in community
  - Part time: continuous input, but a lot of work
  - Short term: inconsistent across project life

- Face-to-face group or individual activities

Degrees of user involvement

- Online contributions from thousands of users
  - Online Feedback Exchange (OFE) systems
  - Crowdsourcing design ideas
  - Citizen science

- User involvement after product release

What is a User-centered Approach?

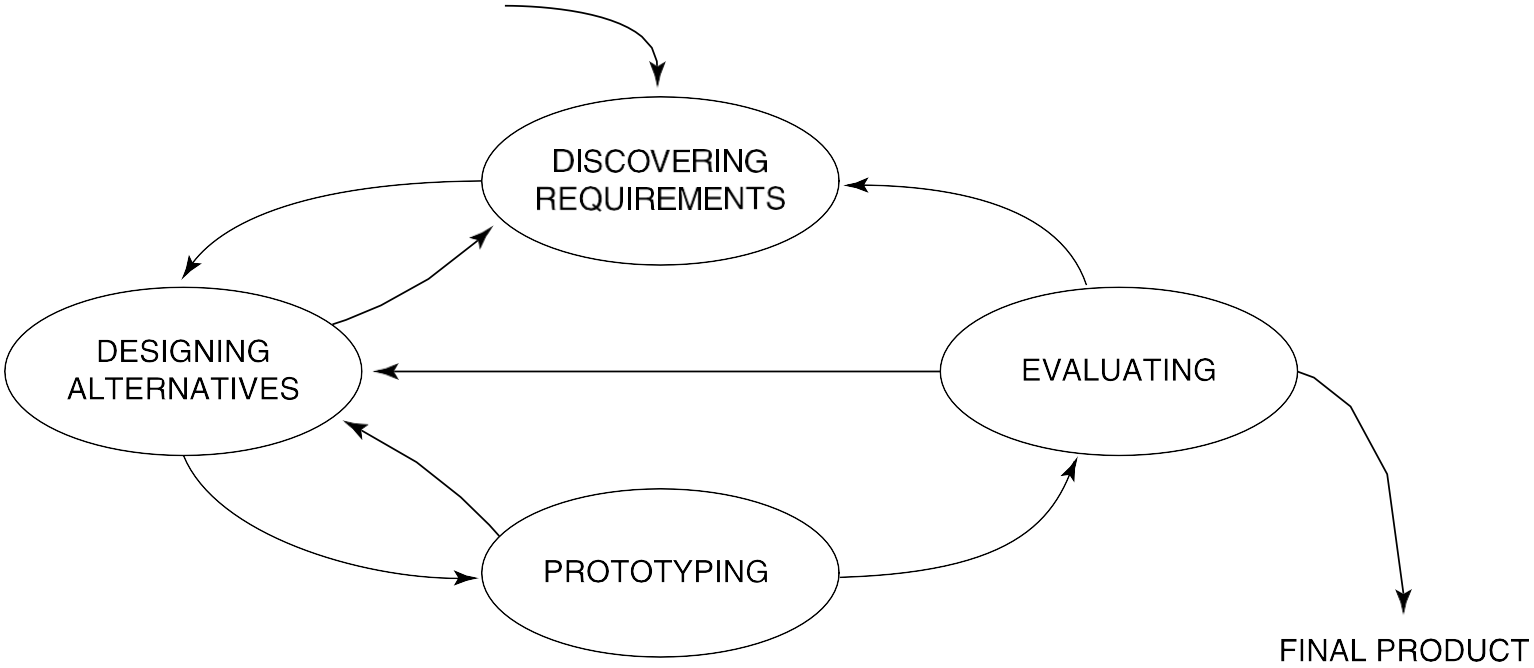
- **Early focus on users and tasks:** directly studying cognitive, behavioral, anthropomorphic, and attitudinal characteristics

- **Empirical measurement:** users' reactions and performance to scenarios, manuals, simulations, and prototypes are observed, recorded, and analyzed

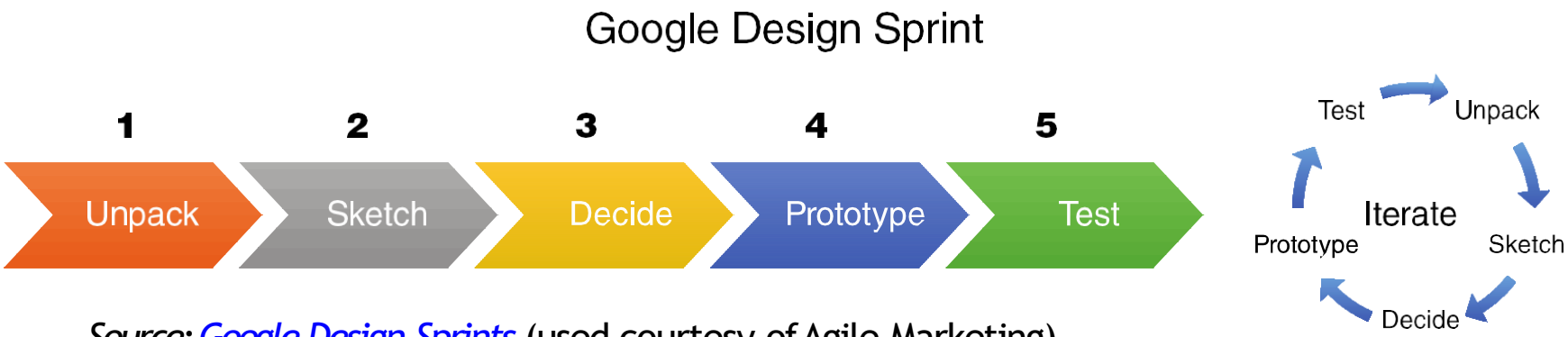
- **Iterative design:** when problems are found in user testing, fix them and carry out more tests

Four Activities of Interaction Design

Exemplifies a user-centered design approach



Google Design Sprints (Knapp et al., 2016)



Source: *Google Design Sprints* (used courtesy of Agile Marketing)

Some practical issues

- Who are the users?
- What are the users' needs?
- How to generate alternative designs?
- How to choose among alternatives?
- How to integrate interaction design activities with other lifecycle models?

Who are the users/stakeholders?

Not obvious

- 382 distinct types of users for smartphone apps (Sha Zhao et al, 2016)
- Many products are intended for use by large sections of the population, so user is "everybody"
- More targeted products are associated with specific roles

Stakeholders

- Identifying stakeholders helps identify groups to include in interaction design activities

What are the users' needs?

- Users rarely know what is possible, instead:
  - **Explore** the problem space
  - **Investigate** user activities to see what can be improved
  - **Try out ideas** with potential users

How to generate alternatives

- Where do alternative designs come from?
  - ‘Flair and creativity’: research and synthesis
  - Cross-fertilization of ideas from different perspectives
  - Users can generate different designs
  - Seek inspiration: similar products and domain, or different products and domain

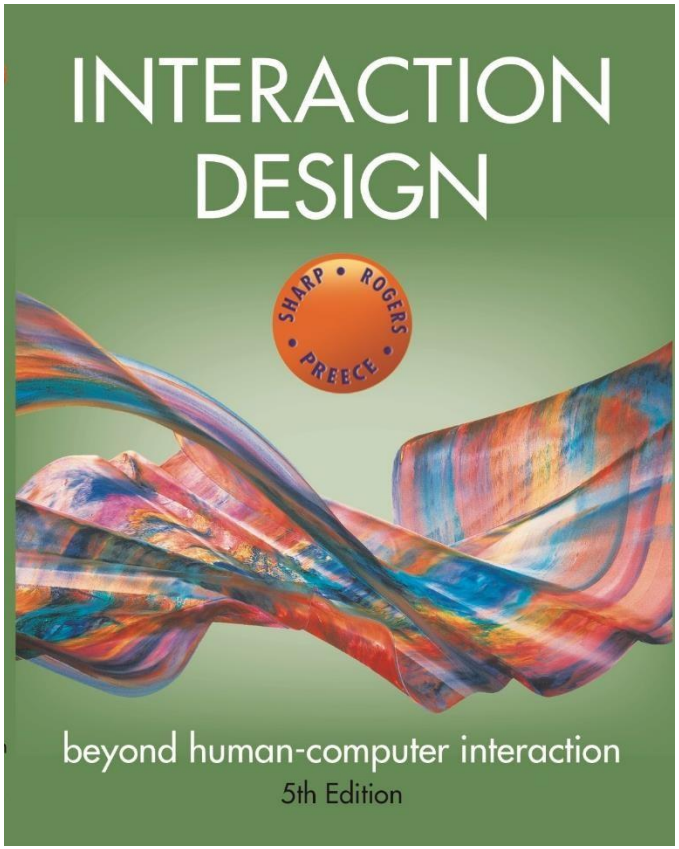
How to generate alternatives

How to choose among alternatives

- Technical feasibility
- Evaluation with users or peers
  - Prototypes not static documentation because behavior is key
- A/B Testing
  - Online method to inform the choice between alternatives
  - Nontrivial to set appropriate metrics and choose user group sets
- Quality thresholds
  - Different stakeholder groups have different quality thresholds
  - Usability and user experience goals lead to relevant criteria

Integrate Design Activities

- Integrating interaction design activities in lifecycle models from other disciplines requires careful planning
- Software development lifecycle models are prominent
- Integrating with agile software development is promising because:
  - It incorporates tight iterations
  - It champions early and regular feedback
  - It handles emergent requirements
  - It aims to strike a balance between flexibility and structure

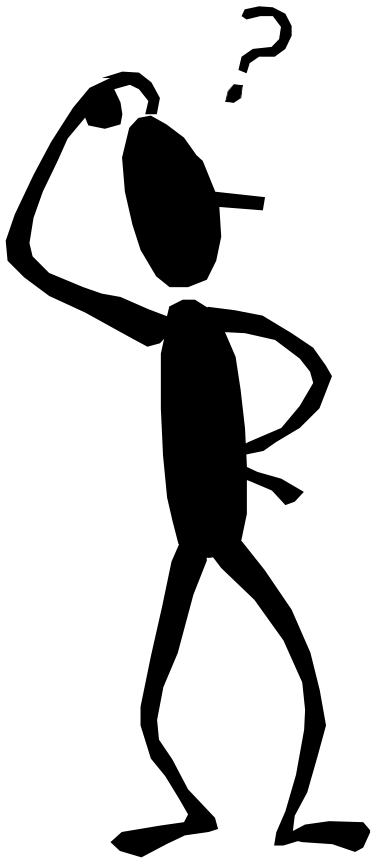


Chapter 13

INTERACTION DESIGN IN PRACTICE

Overview

- AgileUX
- Design Patterns
- Open-Source Resources
- Tools for Interaction Design



Agile development

- Short (one to three week) timeboxes of iterative development (sprint, iteration, cycle)
- Early and repeated customer/user feedback
- Re-prioritization of work based on customer or user so that emergent requirements can be handled
- Many approaches, for example, eXtreme Programming (XP), Scrum, and Kanban.

Technical Debt in UX

- Making compromises that are expedient in short term, but cause problems in long term
- Refactoring to correct pragmatic trade-offs
- UX debt arises from trade-offs
  - A history of neglecting UX
  - Large portfolio of products developed independently

AgileUX

- Integrates techniques from interaction design and agile methods
- ID and Agile have common characteristics such as iteration, user-involvement, and focus on a measurable common criteria
- In Agile iterations, requirements are elaborated and re-prioritized, rather than specified up-front

AgileUX *(continued)*

- All techniques in UX are still relevant, but when to use and how much needs careful planning
  - Focus on the product, not design, as deliverable
  - Cross-functional teams

- Three practical areas: user research, aligning work practices, and documentation

User Research

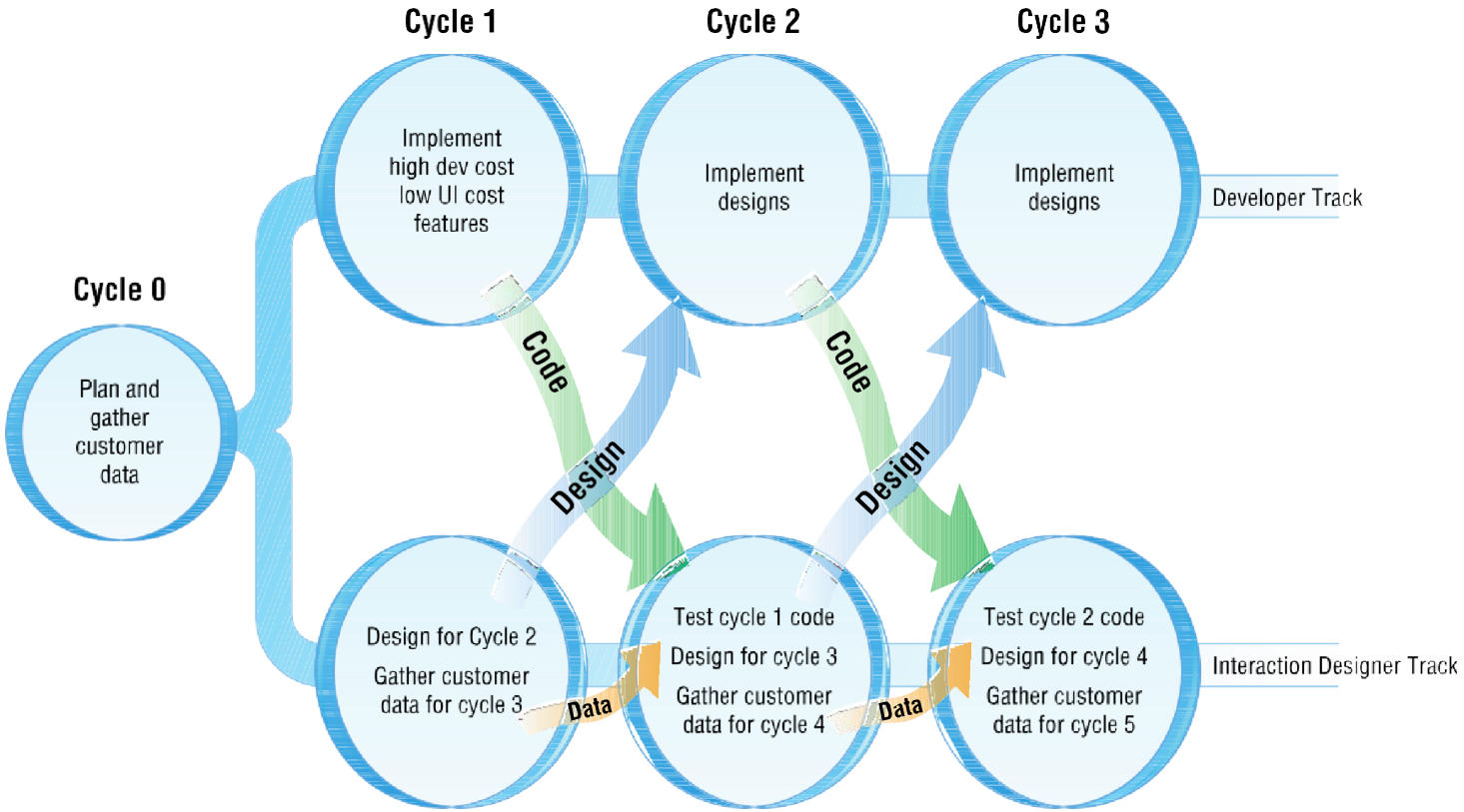
- Seeks to characterize users, tasks, and context through data gathering and analysis

- Detailed user research cannot be fitted within a limited timebox
  - Some user research can be performed in iteration 0 (zero), before implementation starts
  - Ongoing program of user research over a longer period of time
  - Example = LeanUX

Aligning work practices

- Designing a complete product upfront causes problems because of re-prioritization
- Some upfront work is needed (technical and UX)
- Use a parallel tracks approach:
  - Create product vision before development starts
  - Do design work one iteration ahead of development
  - Some teams work two iterations ahead

Parallel tracks approach to AgileUX



Source: Sy, D. (2007) Adapting usability investigations for development, *Journal of Usability* Studies 2(3), May, 112–130. User Experience Professionals Association.

Travel Organizer Example with Agile

1. As a <group traveler>, I want <to choose from a range of potential vacations that suit the group's preferences> so that <the whole group can have a good time>.
2. As a <group traveler>, I want <to know the visa restrictions for everyone in the group> so that <visas can be arranged for everyone in the group in plenty of time>.
3. As a <group traveler>, I want <to know the vaccinations required to visit the chosen destination> so that <vaccinations can be arranged for everyone in the group in plenty of time>.
4. As a <travel agent>, I want <up-to-date information displayed> so that <my clients receive accurate information>.

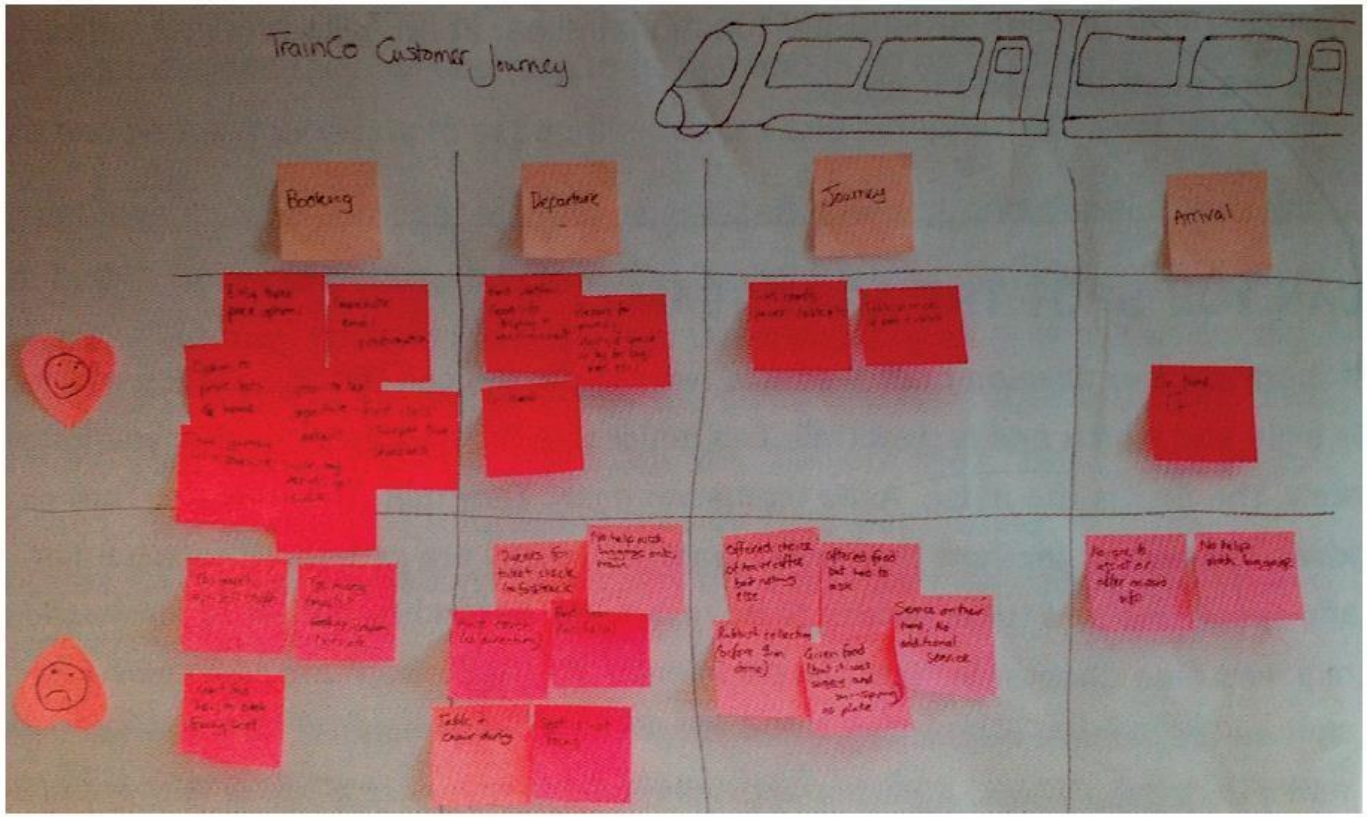
1. Prioritize the main activities
2. Design them and send them to the development team
3. Postpone the reqs and design of unprioritized activities until a product which allows users to choose a vacation is delivered
4. After delivery, customer can decide for unprioritized activities

Documentation

- Common communication approach for UX designers
- Agile encourages minimal documentation so more time can be spent on design and discussion
- Only use documentation where needed. Ask:
  - Can the amount of time spent on documentation be reduced?
  - Who uses documentation?
  - What is the minimum needed by customers?
  - How efficient is the sign-off process?
  - Is there duplication anywhere?
  - How polished does documentation need to be?

Documentation: How polished?

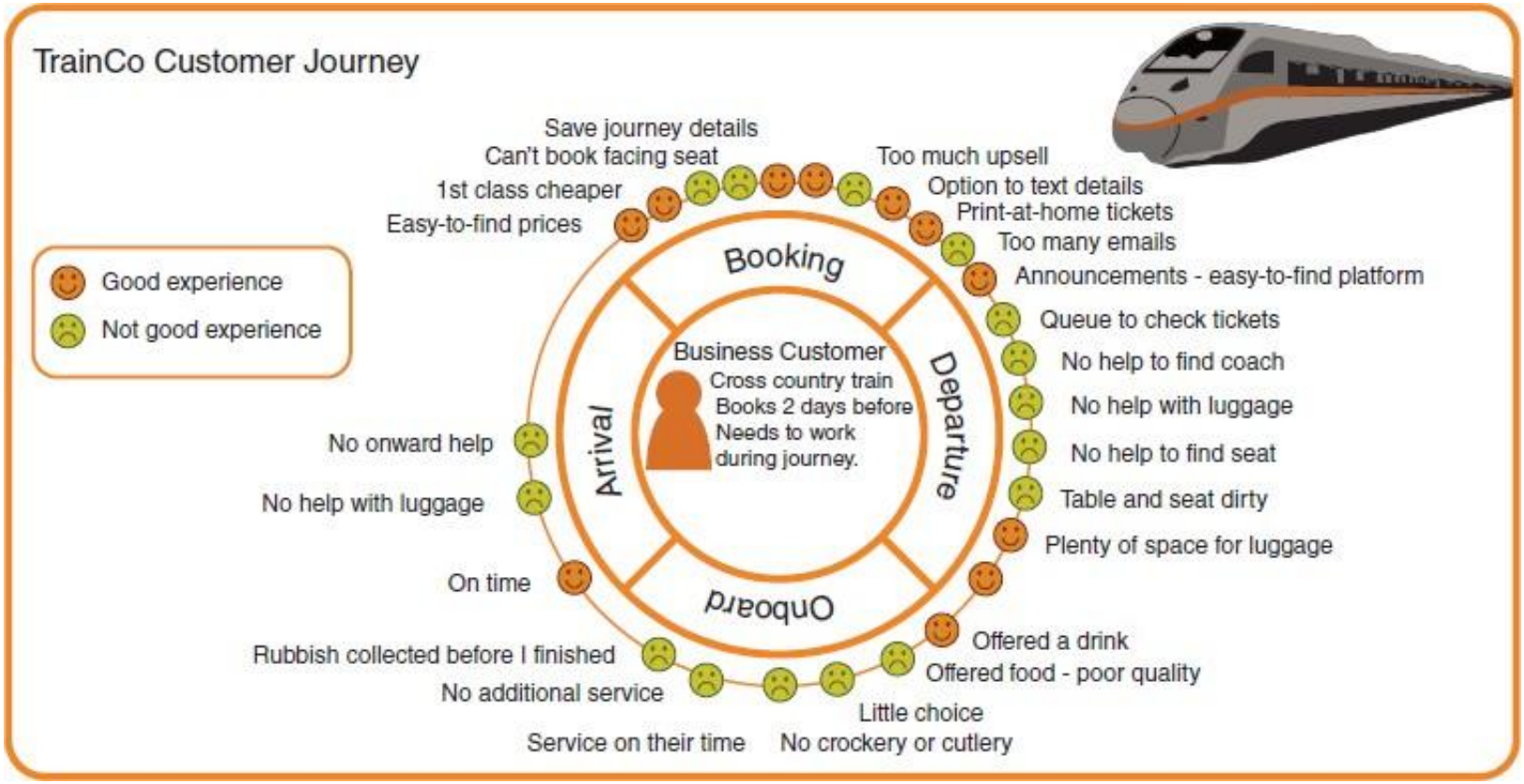
(low fidelity)



Source: Ratcliffe, L. and McNeill, M. (2012) Agile Experience Design. New Riders.

Documentation: how polished?

(high fidelity)



Source: Ratcliffe, L. and McNeill, M. (2012) Agile Experience Design. New Riders.

Design Patterns (DPs)

- Principles and idiomatic solutions built by experienced OO developers to guide in the creation of software.
- In OO Design, a pattern is a named description of a problem and solution that can be applied to new contexts
- Ideally a pattern advises on how to apply its solution in varying circumstances and considers the forces and trade-offs
- Patterns often are associated with software components, for example, GitHub or platform websites

DP Example: High Cohesion

Problem

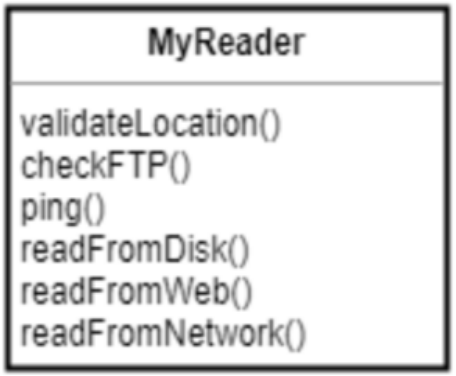
-How to keep complexity manageable?

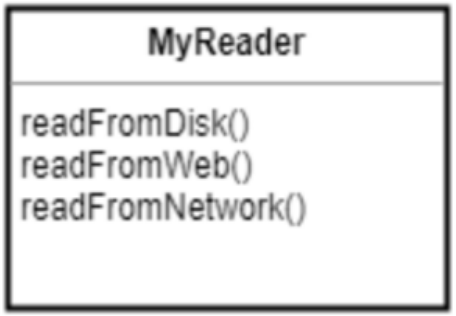
Solution

-If elements are used together then keep them together

Benefits

-Easy to manage
-Easy to understand
-Supports reuse





Design Patterns (DPs)

- Best practice : Use patterns each time it's possible.
- Bad practice : Use patterns everywhere and for everything => Lead to over architecture or unbalanced and inefficient code
- Patterns generally lead to:
  - More general design, More adaptability
  - Less maintainable architecture
- You have to analyze and always find the good tradeoff between genericity and specialization.

Open-Source Resources

- Components, frameworks, systems available free of charge
- Community-driven
- Available for interaction design:
  - Design pattern libraries https://python-patterns.guide/
  - Bootstrap framework

Tools for Interaction Design

- Tools support all aspects of the design process:
  - Creativity, sketching, simulation, brainstorming, library search, mind mapping, and video capture
- Tools integrate together to speed up prototyping
- Tooling landscape changes all the time
- Interactive wireframes or mockups can be produced
- Higher-fidelity prototypes can be produced by linking interactive wireframes to design pattern library with software components